

MEACHAM, OR, USA

WMO: 726885

Lat: 45.511N

Lon: 118.425W

Elev: 3723

StdP: 12.82

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24152

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.8	5.7	-7.5	4.2	-1.3	0.6	6.5	7.2	10.8	37.9	9.5	35.9	0.2	230	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.4	87.8	59.8	84.5	58.9	81.5	58.2	62.7	81.2	61.0	79.8	59.5	77.8	4.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.1	77.0	66.5	53.8	70.8	64.8	51.7	65.4	64.4	30.0	80.7	28.7	79.9	27.5	78.0	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	6.9	6.4	-15.0	93.3	8.2	3.4	-20.9	95.8	-25.7	97.7	-30.3	99.6	-36.2	102.1		
(5)			-15.5	66.6	8.1	3.1	-21.3	68.8	-26.1	70.6	-30.6	72.3	-36.5	74.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	29.2	30.0	34.8	40.3	47.5	53.3	60.5	58.9	52.1	42.7	34.6	27.3		
	DBStd	13.45	8.70	9.05	7.00	6.15	6.61	6.37	5.48	5.84	6.74	7.05	9.01	9.18		
	HDD50	3605	646	559	471	296	130	34	2	4	52	244	464	703		
	HDD65	8186	1111	979	936	740	544	356	158	201	390	692	911	1168		
	CDD50	933	0	0	1	6	51	132	328	281	114	17	3	0		
	CDD65	39	0	0	0	0	0	4	19	14	2	0	0	0		
	CDH74	3113	0	0	0	2	68	244	1293	1124	372	10	0	0		
	CDH80	963	0	0	0	0	10	62	431	364	96	0	0	0		
	WSAvg	1.7	1.6	1.6	1.8	1.9	2.0	2.0	1.8	1.7	1.5	1.3	1.5	1.6		
	PrecAvg	27.3	3.4	2.4	2.9	2.7	2.6	1.9	0.6	0.6	0.9	2.1	3.6	3.5		
(15) Precipitation	PrecMax	34.4	4.8	4.3	4.1	4.3	5.7	3.8	2.4	2.1	2.8	3.5	6.9	6.3		
	PrecMin	21.7	1.8	1.2	1.2	1.5	0.6	0.5	0.1	0.1	0.2	1.0	1.1	2.0		
	PrecStd	3.5	0.9	0.8	0.8	0.8	1.2	0.7	0.5	0.6	0.6	0.7	1.3	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	54.7	62.0	72.0	80.8	86.8	92.0	91.7	87.7	75.0	61.4	49.2		
		MCWB	41.0	43.4	47.0	53.4	57.8	61.5	62.1	61.0	57.4	54.5	47.8	42.5		
	2%	DB	44.8	48.7	56.2	65.6	74.9	80.8	88.0	87.6	82.5	68.1	55.6	44.2		
		MCWB	38.2	40.7	43.7	51.0	55.7	59.5	60.5	59.4	56.6	50.9	46.0	39.3		
	5%	DB	41.7	44.9	51.4	60.1	70.1	76.0	84.9	84.1	78.5	63.2	51.4	41.2		
		MCWB	36.7	38.6	41.8	47.3	53.9	57.1	59.5	58.5	55.4	49.0	44.4	37.1		
	10%	DB	39.2	41.8	47.5	54.7	64.8	71.7	81.6	80.7	73.5	57.8	47.5	38.4		
		MCWB	34.9	36.6	39.8	44.4	51.3	55.1	58.7	57.7	53.9	46.8	41.8	35.2		
	0.4%	WB	42.4	44.9	48.6	55.3	60.1	64.8	66.5	64.2	61.6	56.1	51.1	43.2		
		MCDB	46.1	51.2	59.0	67.7	75.9	82.2	83.1	84.8	78.3	69.9	57.4	47.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.5	41.9	45.1	51.5	57.4	60.8	63.2	61.8	58.9	52.7	47.7	40.6		
		MCDB	43.4	47.4	53.1	63.9	71.6	77.2	81.4	82.3	77.1	64.0	53.8	43.5		
	5%	WB	37.3	39.5	42.7	48.2	54.9	58.2	61.4	60.2	56.7	50.3	45.0	37.7		
		MCDB	40.5	43.9	50.1	58.4	67.1	73.6	80.4	79.5	74.3	60.9	50.5	40.2		
	10%	WB	35.4	37.1	40.7	45.3	52.5	56.0	59.8	58.6	54.6	48.1	42.3	35.4		
		MCDB	38.5	40.7	46.9	53.7	63.0	69.3	78.8	77.5	71.5	56.6	47.1	37.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	17.5	20.0	22.2	25.3	27.7	36.4	37.9	34.5	25.5	16.4	14.3		
		MCDBR	16.3	21.0	27.8	34.3	37.5	37.8	42.7	44.4	45.4	36.3	21.9	14.6		
		MCWBR	11.3	14.0	17.7	20.9	20.2	18.3	18.3	19.5	22.0	21.0	14.0	10.7		
		MCDBR	14.4	18.0	24.7	32.3	32.3	34.2	37.8	39.1	39.2	31.2	20.0	12.9		
	5% WB	MCWBR	10.6	12.5	16.4	20.3	18.1	17.5	17.9	18.6	20.2	18.8	13.5	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.272	0.284	0.322	0.330	0.325	0.327	0.329	0.315	0.293	0.274	0.265		
	taud		2.520	2.564	2.567	2.439	2.445	2.484	2.491	2.472	2.497	2.562	2.564	2.527		
	Ebn at Noon		272	295	305	298	298	298	296	292	288	282	270	263		
	Edh at Noon		20	23	27	34	35	34	33	32	29	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		416	712	1106	1619	1962	2263	2445	2067	1529	943	506	333		
	RadStd		43	85	94	111	159	121	85	116	98	71	41	39		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page